7.Write a program to implement Logistic Regression (LR) algorithm in python

CODE:

```python
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
# Load Breast Cancer dataset
data = load_breast_cancer()
X = data.data
y = data.target
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)
# Scale the data
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
# Logistic Regression Model
model = LogisticRegression(max_iter=5000)
# Train the model
model.fit(X_train, y_train)
# Predict
y_pred = model.predict(X_test)
# Accuracy
print("Predicted Values:")
print(y_pred)
print("\nActual Values:")
print(y_test)
print("\nAccuracy:", accuracy_score(y_test, y_pred) * 100)
```

OUTPUT:

```
Predicted Values:
[1 0 1 0 0 0 0 0 1 1 1 0 0 1 1 1 1 1 1 0 1 1 0 1 0 1 1 0 0 0 0 1 0 0 1 1 0
 1 0 1 1 1 1 1 1 0 1 1 1 0 0 0 1 1 1 1 1 0 1 1 1 0 1 1 1 1 1 0 1 1 1 1 1 0
 1 0 0 1 1 0 1 0 1 0 1 1 0 1 0 1 1 0 1 1 0 0 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0
 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 0 0 0 1 1 1 0 1 0 0 1 1 0 0 0 1 0 0 0 1 1
 1 0 1 1 1 0 1 1 1 1 1 1 1 1 1 0 1 1 0 0 0 1 1]

Actual Values:
[1 0 1 0 0 0 0 0 1 1 1 0 0 1 1 1 1 1 1 0 1 1 0 1 0 1 1 0 0 0 0 1 0 0 1 1 0
 1 0 1 1 1 1 1 1 0 1 1 1 0 0 0 1 1 1 1 1 0 1 1 1 0 1 1 1 1 1 0 1 1 1 1 0 0
 1 0 0 0 1 0 1 0 1 0 1 1 0 1 0 1 1 0 1 1 0 0 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0
 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 0 0 1 1 1 0 1 0 0 1 1 1 0 0 1 0 1 0 1 1
 1 0 1 1 1 0 1 1 1 1 1 1 1 1 1 0 1 1 0 0 0 1 1]

Accuracy: 97.07602339181285
```